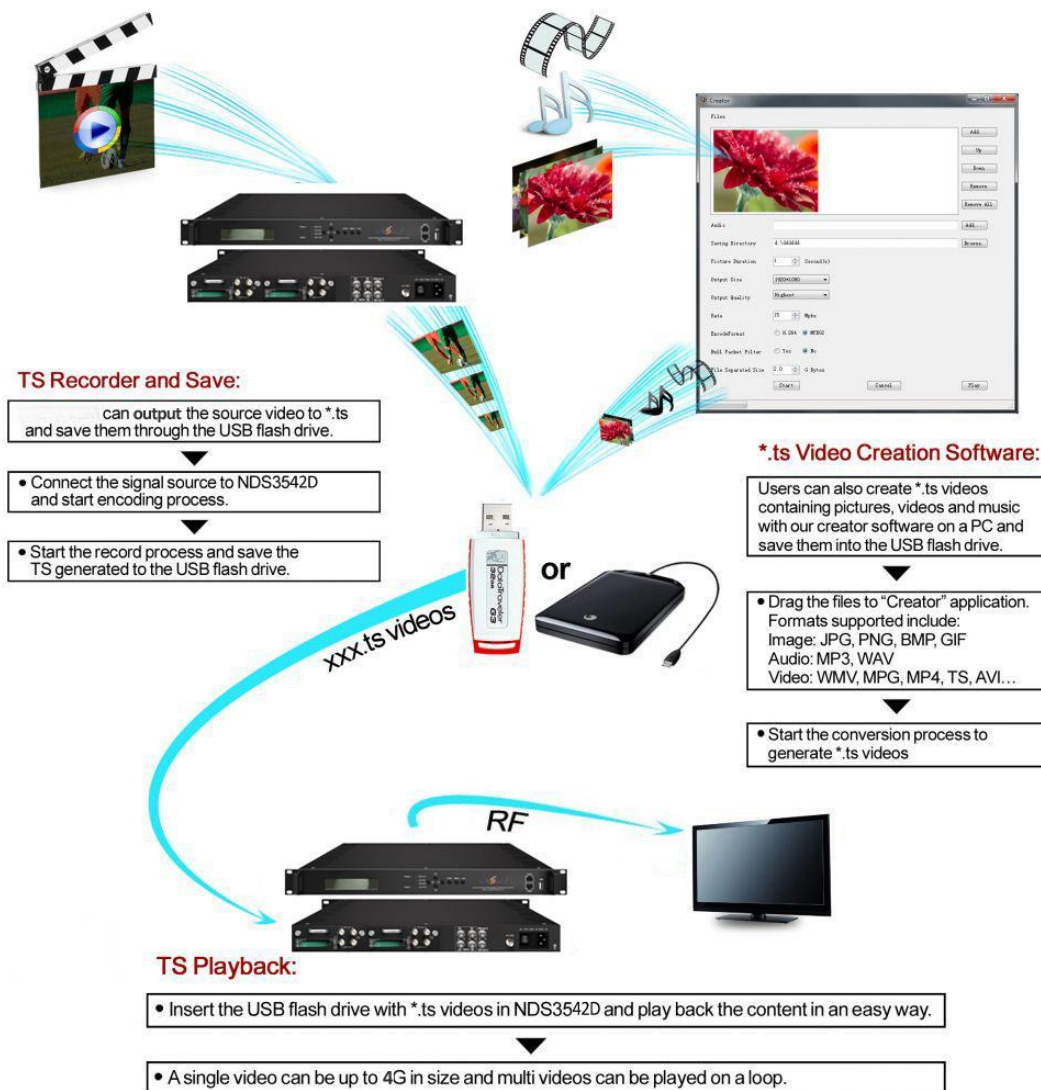


FMUSER FUQ-4653E Modulator 4 Tuner + 2 ASI Inputs
DVB-T-T2-S-S2-S2X to DVB-C RF ASI UDP IP Outputs Integrated
Demodulation Transmux Modulation for Broadcast
Headends Product overview



The FMUSER FUQ-4653E modulator integrates demodulation, trans-mux, and modulation in a 1-U case, supporting 4 tuner inputs for satellite/cable/terrestrial sources and 2 ASI inputs, with outputs including RF, ASI, and UDP IP ports. It processes signals by converting them for distribution, featuring an LCD display, keyboard, and Web NMS for control via web or USB. Ideal for broadcasting headends, cable TV systems, and IP-based content distribution networks.

USB Recoder & Player

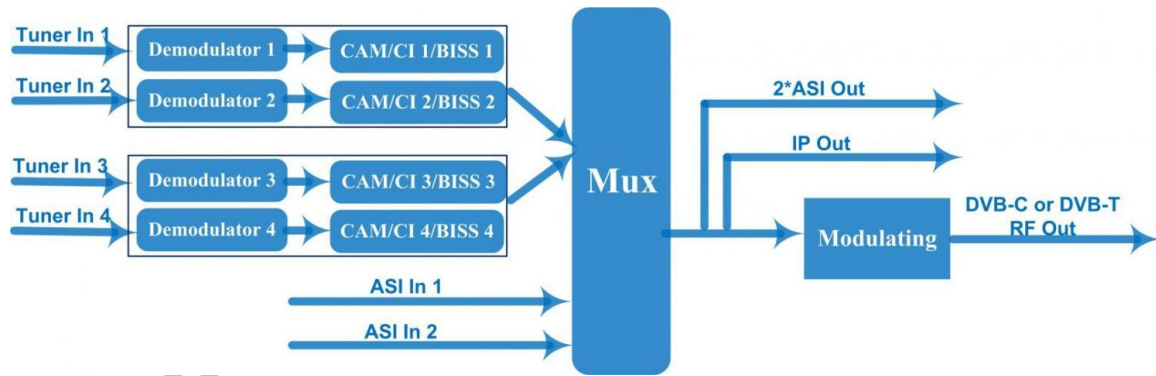


USB Flash Drive Specifications Required: High Speed 2.0; File System FAT32

Key feature

Delivers RF outputs (e.g., DVB-T or DVB-C), ASI outputs, and UDP IP outputs (MPTS or SPTS)

Working Principle



Specifications

Input	4 Tuner (DVB-T/(T)/S/S2/S2 X optional) and 2 ASI input	4 Tuner (DVB-T/(T)/S/S2/S2 X optional) and 2 ASI input	4 Tuner (DVB-T/(T)/S/S2/S2 X optional) and 2 ASI input	4 Tuner (DVB-T/(T)/S/S2/S2 X optional) and 2 ASI input	4 Tuner (DVB-T/(T)/S/S2/S2 X optional) and 2 ASI input	4 Tuner (DVB-T/(T)/S/S2/S2 X optional) and 2 ASI input	4 Tuner (DVB-T/(T)/S/S2/S2 X optional) and 2 ASI input
Tuner Section	DVB-T/(T)	DVB-T/(T)	DVB-T/(T)	Input Frequency	30~1000 MHz	30~1000 MHz	30~1000 MHz
Tuner Section	DVB-T/(T)	DVB-T/(T)	DVB-T/(T)	Bandwidth	6/7/8M	6/7/8M	6/7/8M
Tuner Section	Version 1	DVB-S	DVB-S	Input Frequency	950~2150MHz	950~2150MHz	950~2150MHz
Tuner Section	Version 1	DVB-S	DVB-S	Symbol rate	1~45 Msps	1~45 Msps	1~45 Msps
Tuner Section	Version 1	DVB-S	DVB-S	Signal Strength	-65~-25dBm	-65~-25dBm	-65~-25dBm
Tuner Section	Version 1	DVB-S	DVB-S	FEC	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
Tuner Section	Version 1	DVB-S	DVB-S	Constellation	QPSK	QPSK	QPSK
Tuner Section	Version 1	DVB-S	DVB-S	Max input bitrate	≤120 Mbps	≤120 Mbps	≤120 Mbps
Tuner Section	Version 1	DVB-S2	DVB-S2	Input Frequency	950~2150MHz	950~2150MHz	950~2150MHz

Tuner Section	Version 1	DVB-S2	DVB-S2	Symbol rate	QPSK 1~45 Mps 8PSK 2~30 Mps	QPSK 1~45 Mps 8PSK 2~30 Mps	QPSK 1~45 Mps 8PSK 2~30 Mps
Tuner Section	Version 1	DVB-S2	DVB-S2	FEC	1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10	1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10	1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10
Tuner Section	Version 1	DVB-S2	DVB-S2	Constellation	QPSK, 8PSK	QPSK, 8PSK	QPSK, 8PSK
Tuner Section	Version 1	DVB-S2	DVB-S2	Max input bitrate	≤120 Mbps	≤120 Mbps	≤120 Mbps
Tuner Section	Version 2 (New)	DVB-S	DVB-S	Input Frequency	950~215 0MHz	950~215 0MHz	950~215 0MHz
Tuner Section	Version 2 (New)	DVB-S	DVB-S	Symbol rate	0.5~45Mps	0.5~45Mps	0.5~45Mps
Tuner Section	Version 2 (New)	DVB-S	DVB-S	Signal Strength	- 65- -25dBm	- 65- -25dBm	- 65- -25dBm
Tuner Section	Version 2 (New)	DVB-S	DVB-S	FEC	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
Tuner Section	Version 2 (New)	DVB-S	DVB-S	Constellation	QPSK	QPSK	QPSK
Tuner Section	Version 2 (New)	DVB-S	DVB-S	Max input bitrate	≤160 Mbps	≤160 Mbps	≤160 Mbps
Tuner Section	Version 2 (New)	DVB-S2	DVB-S2	Input Frequency	950~215 0MHz	950~215 0MHz	950~215 0MHz
Tuner Section	Version 2 (New)	DVB-S2	DVB-S2	Symbol rate	QPSK/8PSK /16APSK: 0.5~45 Mps 32APSK: 0.5~40Mps;	QPSK/8PSK /16APSK: 0.5~45 Mps 32APSK: 0.5~40Mps;	QPSK/8PSK /16APSK: 0.5~45 Mps 32APSK: 0.5~40Mps;

Tuner Section	Version 2 (New)	DVB-S2	DVB-S2	FEC	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10
					8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10	8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10	8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10
					16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10	16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10	16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10
					32APSK: 3/4, 4/5, 5/6, 8/9, 9/10	32APSK: 3/4, 4/5, 5/6, 8/9, 9/10	32APSK: 3/4, 4/5, 5/6, 8/9, 9/10
Tuner Section	Version 2 (New)	DVB-S2	DVB-S2	Constellation	QPSK, 8PSK, 16APSK, 32APSK	QPSK, 8PSK, 16APSK, 32APSK	QPSK, 8PSK, 16APSK, 32APSK
Tuner Section	Version 2 (New)	DVB-S2	DVB-S2	Max input bitrate	≤160 Mbps	≤160 Mbps	≤160 Mbps
Tuner Section	Version 2 (New)	DVB-S2X	DVB-S2X	Input Frequency	950~2150MHz	950~2150MHz	950~2150MHz
Tuner Section	Version 2 (New)	DVB-S2X	DVB-S2X	Symbol rate	QPSK/8PSK /16APSK :0.5~45 Msps 8A PSK/32A PSK: 0.5 ~40Msps	QPSK/8PSK /16APSK :0.5~45 Msps 8A PSK/32A PSK: 0.5 ~40Msps	QPSK/8PSK /16APSK :0.5~45 Msps 8A PSK/32A PSK: 0.5 ~40Msps

Tuner Section	Version 2 (New)	DVB-S2X	DVB-S2X	FEC	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 8APSK: 5/9-L, 26/45-L 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36 , 2/3-L, 25/36, 13/18, 7/9, 77/90 32APSK: 3/4, 4/5, 5/6, 8/9, 2/3-L, 32/45, 11/15, 7/9	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 8APSK: 5/9-L, 26/45-L 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36 , 2/3-L, 25/36, 13/18, 7/9, 77/90 32APSK: 3/4, 4/5, 5/6, 8/9, 2/3-L, 32/45, 11/15, 7/9	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 8APSK: 5/9-L, 26/45-L 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36 , 2/3-L, 25/36, 13/18, 7/9, 77/90 32APSK: 3/4, 4/5, 5/6, 8/9, 2/3-L, 32/45, 11/15, 7/9
Tuner Section	Version 2 (New)	DVB-S2X	DVB-S2X	Constellation	QPSK, 8PSK, 8APSK, 16APSK, 32APSK	QPSK, 8PSK, 8APSK, 16APSK, 32APSK	QPSK, 8PSK, 8APSK, 16APSK, 32APSK
Tuner Section	Version 2 (New)	DVB-S2X	DVB-S2X	Max input bitrate	≤160 Mbps	≤160 Mbps	≤160 Mbps

Multiplexing	Maximum PID Remapping	Maximum PID Remapping	Maximum PID Remapping	128per input channel	128per input channel	128per input channel	128per input channel
Multiplexing	Function	Function	Function	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)
Multiplexing	Function	Function	Function	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting
Multiplexing	Function	Function	Function	Generate PSI/SI table automatically	Generate PSI/SI table automatically	Generate PSI/SI table automatically	Generate PSI/SI table automatically
Modulation	DVB-C	DVB-C	DVB-C	Standard	J.83A(DVB-C), J.83B, J.83C	J.83A(DVB-C), J.83B, J.83C	J.83A(DVB-C), J.83B, J.83C
Modulation	DVB-C	DVB-C	DVB-C	MER	≥40dB	≥40dB	≥40dB
Modulation	DVB-C	DVB-C	DVB-C	RF frequency	30~999MHz, 1KHz step	30~999MHz, 1KHz step	30~999MHz, 1KHz step
Modulation	DVB-C	DVB-C	DVB-C	RF output level	-20~0dbm(87~107dbμV), 0.1db step	-20~0dbm(87~107dbμV), 0.1db step	-20~0dbm(87~107dbμV), 0.1db step
Modulation	DVB-C	DVB-C	DVB-C	Symbol rate	5.0Msps-9.0Msps, 1ksps stepping	5.0Msps-9.0Msps, 1ksps stepping	5.0Msps-9.0Msps, 1ksps stepping
Modulation	DVB-C	DVB-C	DVB-C	RF Out	4*DVB-C carriers output	4*DVB-C carriers output	4*DVB-C carriers output
Modulation	DVB-C	DVB-C	DVB-C		J.83A	J.83B	J.83C
Modulation	DVB-C	DVB-C	DVB-C	Constellation	16/32/64/128/256QAM	64/256 QAM	64/256 QAM

Modulation	DVB-C	DVB-C	DVB-C	Bandwidth	8M	6M	6M
Modulation	DVB-T	DVB-T	DVB-T	Bandwidth	6M, 7M, 8M	6M, 7M, 8M	6M, 7M, 8M
Modulation	DVB-T	DVB-T	DVB-T	Constellation	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
Modulation	DVB-T	DVB-T	DVB-T	Code rate	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
Modulation	DVB-T	DVB-T	DVB-T	Guard Interval	1/32, 1/16, 1/8, 1/4	1/32, 1/16, 1/8, 1/4	1/32, 1/16, 1/8, 1/4
Modulation	DVB-T	DVB-T	DVB-T	Transmission Mode	2K, 8K	2K, 8K	2K, 8K
Modulation	DVB-T	DVB-T	DVB-T	MER	≥40dB	≥40dB	≥40dB
Modulation	DVB-T	DVB-T	DVB-T	RF frequency	30~999M Hz, 1KHz step	30~999M Hz, 1KHz step	30~999M Hz, 1KHz step
Modulation	DVB-T	DVB-T	DVB-T	RF Out	2*DVB-T carriers output	2*DVB-T carriers output	2*DVB-T carriers output
Modulation	DVB-T	DVB-T	DVB-T	RF output level	-20~0dbm(87~107 dbμV), 0.1db step	-20~0dbm(87~107 dbμV), 0.1db step	-20~0dbm(87~107 dbμV), 0.1db step

TS Output	2 ASI out as mirror of 2 RF carriers (BNC type) 2 MPTS over UDP out (RJ45, 100/1000M self-adaption) (for DVB-T RF out) 4 MPTS over UDP out (RJ45, 100/1000M self-adaption) (for DVB-C RF out) -- -version■ 8 SPTS over UDP out (RJ45, 100/1000M self-adaption) (for DVB-C RF out) --- version■	2 ASI out as mirror of 2 RF carriers (BNC type) 2 MPTS over UDP out (RJ45, 100/1000M self-adaption) (for DVB-T RF out) 4 MPTS over UDP out (RJ45, 100/1000M self-adaption) (for DVB-C RF out) -- -version■ 8 SPTS over UDP out (RJ45, 100/1000M self-adaption) (for DVB-C RF out) --- version■	2 ASI out as mirror of 2 RF carriers (BNC type) 2 MPTS over UDP out (RJ45, 100/1000M self-adaption) (for DVB-T RF out) 4 MPTS over UDP out (RJ45, 100/1000M self-adaption) (for DVB-C RF out) -- -version■ 8 SPTS over UDP out (RJ45, 100/1000M self-adaption) (for DVB-C RF out) --- version■	2 ASI out as mirror of 2 RF carriers (BNC type) 2 MPTS over UDP out (RJ45, 100/1000M self-adaption) (for DVB-T RF out) 4 MPTS over UDP out (RJ45, 100/1000M self-adaption) (for DVB-C RF out) -- -version■ 8 SPTS over UDP out (RJ45, 100/1000M self-adaption) (for DVB-C RF out) --- version■	2 ASI out as mirror of 2 RF carriers (BNC type) 2 MPTS over UDP out (RJ45, 100/1000M self-adaption) (for DVB-T RF out) 4 MPTS over UDP out (RJ45, 100/1000M self-adaption) (for DVB-C RF out) -- -version■ 8 SPTS over UDP out (RJ45, 100/1000M self-adaption) (for DVB-C RF out) --- version■	2 ASI out as mirror of 2 RF carriers (BNC type) 2 MPTS over UDP out (RJ45, 100/1000M self-adaption) (for DVB-T RF out) 4 MPTS over UDP out (RJ45, 100/1000M self-adaption) (for DVB-C RF out) -- -version■ 8 SPTS over UDP out (RJ45, 100/1000M self-adaption) (for DVB-C RF out) --- version■	2 ASI out as mirror of 2 RF carriers (BNC type) 2 MPTS over UDP out (RJ45, 100/1000M self-adaption) (for DVB-T RF out) 4 MPTS over UDP out (RJ45, 100/1000M self-adaption) (for DVB-C RF out) -- -version■ 8 SPTS over UDP out (RJ45, 100/1000M self-adaption) (for DVB-C RF out) --- version■
System	Local interface	Local interface	LCD + control buttons	LCD + control buttons	LCD + control buttons	LCD + control buttons	LCD + control buttons
System	Remote management	Remote management	Web NMS	Web NMS	Web NMS	Web NMS	Web NMS
System	Language	Language	English and Chinese	English and Chinese	English and Chinese	English and Chinese	English and Chinese

System	Upgradin g	Upgradin g	Web and USB	Web and USB	Web and USB	Web and USB	Web and USB
General	Dimensio n	Dimensio n	482mm× 300mm× 44.5mm(W*D*H)	482mm× 300mm× 44.5mm(W*D*H)	482mm× 300mm× 44.5mm(W*D*H)	482mm× 300mm× 44.5mm(W*D*H)	482mm× 300mm× 44.5mm(W*D*H)
General	Weight	Weight	3.7kg	3.7kg	3.7kg	3.7kg	3.7kg
General	Temperat ure	Temperat ure	0~45■(O peration) ; -20~80 ■(Storag e)	0~45■(O peration) ; -20~80 ■(Storag e)	0~45■(O peration) ; -20~80 ■(Storag e)	0~45■(O peration) ; -20~80 ■(Storag e)	0~45■(O peration) ; -20~80 ■(Storag e)
General	Power	Power	AC 100V ±1050/60 Hz; AC 2 20V±10% , 50/60HZ	AC 100V ±1050/60 Hz; AC 2 20V±10% , 50/60HZ	AC 100V ±1050/60 Hz; AC 2 20V±10% , 50/60HZ	AC 100V ±1050/60 Hz; AC 2 20V±10% , 50/60HZ	AC 100V ±1050/60 Hz; AC 2 20V±10% , 50/60HZ
General	Consump tion	Consump tion	25W	25W	25W	25W	25W

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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